How can we work with "High-dimensional data"?

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Introduction

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- However, the true challenge nowadays is not only to collect, but also to store, manage, and explore this huge quantity of data in order to extract the right insights for decision-making.
- Unfortunately, this valorization is limited by many problems, such as heterogeneity, security, and the **curse of dimensionality** (Bellman, 1961).

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Question 1

Are we still able to easily visualize our data if the dimensionality increases?

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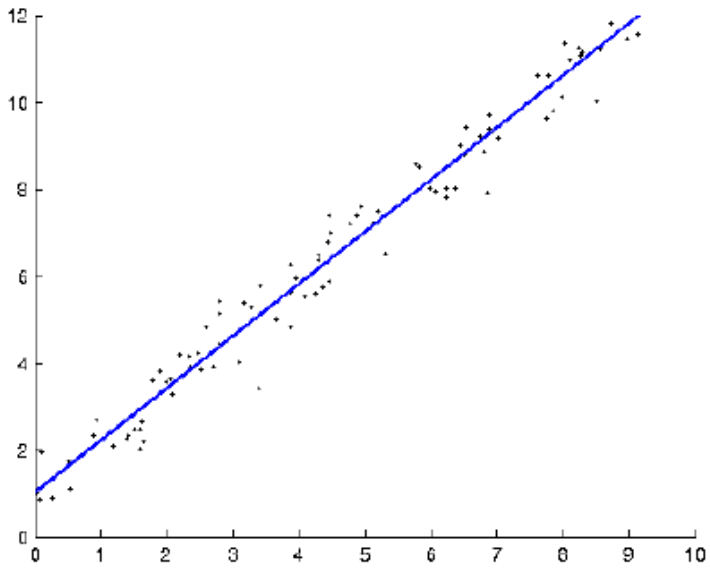
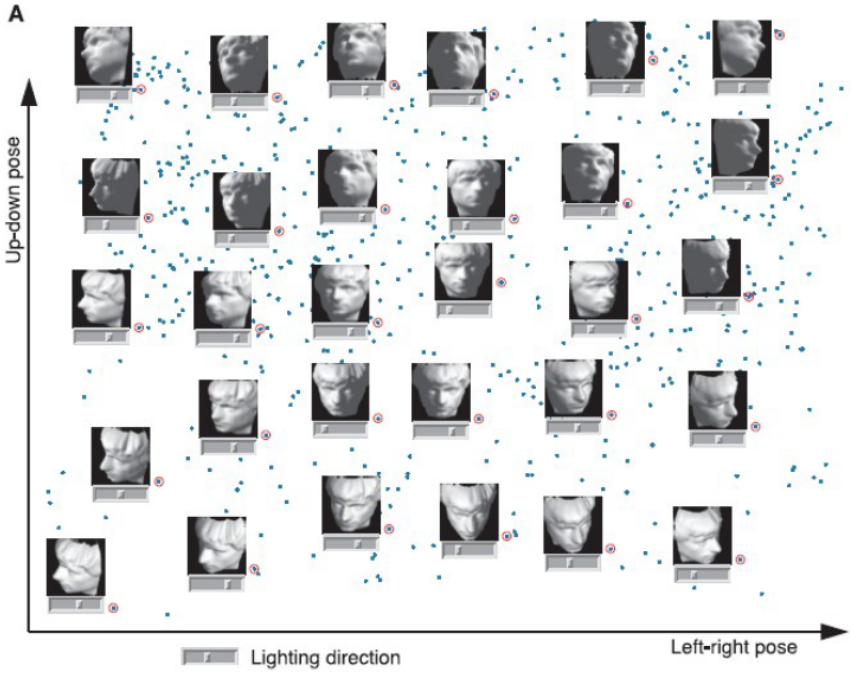


(a) 2D space



(b) 3D space (Tenenbaum et al., 2000)

Figure 1: Data representation in two differents space.

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Question 3

Can we guarantee the performance of our model if it is trained on high-dimensional data, where some features may be redundant and not significant?

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Answer

We need to reduce the dimensionality.

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How can we effectively find this substructure?

- Discovering this underlying structure is the main idea of dimensionality reduction.

References I

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Thank you for your attention

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